Chapter 8. Closed Forms and Exact Forms

The Poincaré lemma and the de Rham groups of punctured euclidean space

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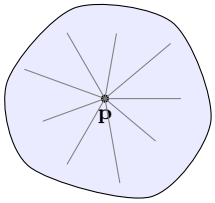
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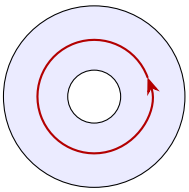
In physics one often asks whether a vector field F in $\mathbb{R}^3$ is the gradient of a scalar field f (F is **conservative**, and f, or its negative, is a **potential function** for F): in the language of forms, whether a 1-form is the differential of a 0-form, i.e. exact. Similarly one asks whether G is the curl of a field F: whether a 2-form is exact. In $\mathbb{R}^n$: if ω is a k-form on an open set A, a necessary condition for exactness is $d\omega = 0$. It is not in general sufficient; this chapter explores what additional conditions on A, or on A and ω, ensure that ω is exact.

§39. The Poincaré Lemma

Motivation. Every closed form is locally exact, and on star-shaped domains it is globally exact. This settles when a vector field has a potential in a convex region, and it is the local fact whose global failure is measured by de Rham cohomology.



star-convex: closed ⇒ exact



punctured: $d\theta$ closed, not exact

Figure. *On a star-convex set every segment from* $\mathbf{p}$ *stays in the set, which is what the Poincaré construction integrates along; there every closed form is exact. Around a hole this fails:* $d\theta$ *is closed on* $\mathbb{R}^2 - \mathbf{0}$ *but its integral around the loop is* $2\pi \neq 0$.

If A is *star-convex*, every closed form on A is exact (the Poincaré lemma). We begin with a preliminary result.

Theorem 39.1 (Leibniz's rule)

Let Q be a rectangle in $\mathbb{R}^n$; let $f : Q \times [a,b] \to \mathbb{R}$ be a continuous function. Denote f by $f(\mathbf{x}, t)$ for $\mathbf{x} \in Q$ and $t \in [a,b]$. Then the function

$$F(\mathbf{x}) = \int_{t=a}^{t=b} f(\mathbf{x}, t)$$

is continuous on Q. Furthermore, if $\partial f/\partial x_j$ is continuous on $Q \times [a,b]$, then

$$\frac{\partial F}{\partial x_j}(\mathbf{x}) = \int_{t=a}^{t=b} \frac{\partial f}{\partial x_j}(\mathbf{x}, t).$$

This is *Leibniz's rule for differentiating under the integral sign.*

Proof.

Step 1. $Q \times [a,b]$ is compact, so f is uniformly continuous: given $\epsilon > 0$ there is $\delta > 0$ with $|f(\mathbf{x}_1, t_1) - f(\mathbf{x}_0, t_0)| < \epsilon$ whenever $|(\mathbf{x}_1, t_1) - (\mathbf{x}_0, t_0)| < \delta$. So for $|\mathbf{x}_1 - \mathbf{x}_0| < \delta$,

$$|F(\mathbf{x}_1) - F(\mathbf{x}_0)| \leq \int_{t=a}^{t=b} |f(\mathbf{x}_1, t) - f(\mathbf{x}_0, t)| \leq \epsilon(b-a).$$

Step 2. Only x_j and t are involved, so we may assume $n = 1$ and $Q = [c,d]$. Let $G(x) = \int_{t=a}^{t=b} D_1 f(x,t)$; we show $F'(x) = G(x)$. Since $D_1 f$ is continuous on $[c,d] \times [a,b]$, Fubini's theorem

and the fundamental theorem of calculus give, for $x_0 \in [c,d]$,

$$\int_{x=c}^{x=x_0} G(x) = \int_{x=c}^{x=x_0}\int_{t=a}^{t=b} D_1 f(x,t) = \int_{t=a}^{t=b}\int_{x=c}^{x=x_0} D_1 f(x,t)$$
$$= \int_{t=a}^{t=b} [f(x_0,t) - f(c,t)] = F(x_0) - F(c).$$

G is continuous by Step 1, so the fundamental theorem of calculus gives $G(x) = F'(x)$. ■

Definition 39.1 (differentiable homotopy)

Let A, B be open in $\mathbb{R}^n$, $\mathbb{R}^m$, and $g, h : A \to B$ of class C^∞. g and h are **differentiably homotopic** if there is a C^∞ map $H : A \times I \to B$ $(I = [0,1])$ with

$$H(\mathbf{x},0) = g(\mathbf{x}) \qquad \text{and} \qquad H(\mathbf{x},1) = h(\mathbf{x}) \qquad (\mathbf{x} \in A).$$

H is a **differentiable homotopy** between g and h.

For each t, $\mathbf{x} \mapsto H(\mathbf{x},t)$ is a C^∞ map $A \to B$; thinking of t as time, H "deforms" g into h as t goes from 0 to 1.

Theorem 39.2

Let A and B be open sets in $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. Let $g, h : A \to B$ be C^∞ maps that are differentiably homotopic. Then there is a linear transformation

$$\mathcal{P} : \Omega^{k+1}(B) \to \Omega^k(A),$$

defined for $k \geq 0$, such that for any form η of order $k > 0$,

$$d\mathcal{P}\eta + \mathcal{P}d\eta = h^*\eta - g^*\eta,$$

while for a form f of order 0,

$$\mathcal{P}df = h^*f - g^*f.$$

Hence if η is a closed form of positive order, $h^*\eta$ and $g^*\eta$ differ by an exact form $(= d\mathcal{P}\eta)$; if f is a closed 0-form, $h^*f = g^*f$. (d raises order by 1 and $\mathcal{P}$ lowers it by 1; $\mathcal{P}f$ is undefined for a 0-form f.)

Proof.

Step 1. A special case. Given A open in $\mathbb{R}^n$, let U be a neighborhood of $A \times I$ in $\mathbb{R}^{n+1}$ and $\alpha, \beta : A \to U$, $\alpha(\mathbf{x}) = (\mathbf{x},0)$, $\beta(\mathbf{x}) = (\mathbf{x},1)$ (differentiably homotopic). We define for each $(k+1)$-form η on U a k-form $P\eta$ on A with

$$dP\eta + Pd\eta = \beta^*\eta - \alpha^*\eta \ \ (\text{order } \eta > 0), \qquad Pdf = \beta^*f - \alpha^*f \ \ (\text{order } f = 0). \qquad (*)$$

Let $\mathbf{x}$ be the general point of $\mathbb{R}^n$ and t that of $\mathbb{R}$; $dx_1, \ldots, dx_n, dt$ are the elementary 1-forms of $\mathbb{R}^{n+1}$. For a continuous scalar function g on $A \times I$ define $(\mathcal{I}g)(\mathbf{x}) = \int_{t=0}^{t=1} g(\mathbf{x},t)$. For $k \geq 0$ write a $(k+1)$-form uniquely as

$$\eta = \sum_{[I]} f_I\, dx_I + \sum_{[J]} g_J\, dx_J \wedge dt,$$

I ascending $(k+1)$-tuples and J ascending k-tuples from $\{1, \ldots, n\}$, and define

$$P\eta = \sum_{[I]} P(f_I\, dx_I) + \sum_{[J]} P(g_J\, dx_J \wedge dt), \qquad P(f_I\, dx_I) = 0, \quad P(g_J\, dx_J \wedge dt) = (-1)^k(\mathcal{I}g_J)\, dx_J.$$

$P\eta$ is a k-form on A; P is linear (uniqueness of the representation and linearity of $\mathcal{I}$), and $\mathcal{I}g$ is C^∞ by Leibniz's rule, so $P\eta$ is C^∞. For $k=0$, $\eta = \sum f_i\,dx_i + g\,dt$, J is empty, and $P\eta = \mathcal{I}g$. (P "integrates η in the direction of the last coordinate".)

Step 2. The formulas $P(f\,dx_I) = 0$ and $P(g\,dx_J \wedge dt) = (-1)^k(\mathcal{I}g)\,dx_J$ hold even for arbitrary (non-ascending) I, J: trivially if indices repeat (then $dx_I = 0$, $dx_J = 0$); by definition if ascending; and in general since rearranging indices changes dx_I, dx_J only by a sign.

Step 3. $(*)$ for $k=0$:

$$P(df) = P\Big(\sum_j \frac{\partial f}{\partial x_j}dx_j\Big) + P\Big(\frac{\partial f}{\partial t}dt\Big) = 0 + (-1)^0\mathcal{I}\Big(\frac{\partial f}{\partial t}\Big) = f\circ\beta - f\circ\alpha = \beta^*f - \alpha^*f,$$

by the fundamental theorem of calculus.

Step 4. $(*)$ for $k>0$. Since $\alpha(\mathbf{x}) = (\mathbf{x},0)$, $\alpha^*(dx_i) = dx_i$ and $\alpha^*(dt) = d(0) = 0$; likewise for β. By linearity it suffices to treat $\eta = f\,dx_I$ and $\eta = g\,dx_J\wedge dt$. For $\eta = f\,dx_I$: the left side is

$$\begin{aligned} dP\eta + Pd\eta &= d(0) + P\Big(\sum_j \frac{\partial f}{\partial x_j}dx_j\wedge dx_I\Big) + P\Big(\frac{\partial f}{\partial t}dt\wedge dx_I\Big)\\ &= 0 + (-1)^{k+1}P\Big(\frac{\partial f}{\partial t}dx_I\wedge dt\Big) = \mathcal{I}\Big(\frac{\partial f}{\partial t}\Big)dx_I = [f\circ\beta - f\circ\alpha]\,dx_I, \end{aligned}$$

using Step 2 ($dt\wedge dx_I = (-1)^{k+1}dx_I\wedge dt$ as dx_I has order $k+1$), and the right side is $\beta^*\eta - \alpha^*\eta = (f\circ\beta)\beta^*(dx_I) - (f\circ\alpha)\alpha^*(dx_I) = [f\circ\beta - f\circ\alpha]\,dx_I$. For $\eta = g\,dx_J\wedge dt$:

$$d(P\eta) = d[(-1)^k(\mathcal{I}g)\,dx_J] = (-1)^k\sum_{j=1}^n D_j(\mathcal{I}g)\,dx_j\wedge dx_J. \qquad (**)$$

On the other hand $d\eta = \sum_j (D_jg)\,dx_j\wedge dx_J\wedge dt + (D_{n+1}g)\,dt\wedge dx_J\wedge dt$, and the last term vanishes, so by Step 2

$$P(d\eta) = (-1)^{k+1}\sum_{j=1}^n \mathcal{I}(D_jg)\,dx_j\wedge dx_J. \qquad (***)$$

By Leibniz's rule $D_j(\mathcal{I}g) = \mathcal{I}(D_jg)$, so $(**)+(***) = 0$. The right side is $\beta^*(g\,dx_J\wedge dt) - \alpha^*(g\,dx_J\wedge dt) = 0$, since $\beta^*(dt) = \alpha^*(dt) = 0$.

Step 5. General case. Let $H : A\times I \to B$ be a differentiable homotopy between g and h, $\alpha,\beta : A\to A\times I$ the maps of Step 1, and P the corresponding transformation. Define $\mathcal{P}\eta = P(H^*\eta)$; since $H^*\eta$ is a $(k+1)$-form on a neighborhood of $A\times I$, $\mathcal{P}\eta$ is a k-form on A. As $H\circ\alpha = g$ and $H\circ\beta = h$, for $k>0$

$$d\mathcal{P}\eta + \mathcal{P}d\eta = dP(H^*\eta) + P(H^*d\eta) = dP(H^*\eta) + P(dH^*\eta) = \beta^*(H^*\eta) - \alpha^*(H^*\eta) = h^*\eta - g^*\eta,$$

by Step 1; the case $k=0$ is similar. ■

Definition 39.2 (star-convex)

An open set A in $\mathbb{R}^n$ is **star-convex** with respect to the point $\mathbf{p}$ of A if for each $\mathbf{x}\in A$ the line segment joining $\mathbf{x}$ and $\mathbf{p}$ lies in A.

Example 39.1

A star-shaped region may be star-convex with respect to one of its points $\mathbf{p}$ but not with respect to another point $\mathbf{q}$. A convex set is star-convex with respect to each of its points. A "C"-shaped region is not star-convex with respect to any of its points.

Theorem 39.3 (The Poincaré lemma)

Let A be a star-convex open set in $\mathbb{R}^n$. If ω is a closed k-form on A, then ω is exact on A.

Proof. Let A be star-convex with respect to $\mathbf{p}$. Let $h : A \to A$ be the identity and $g : A \to A$ the constant map to $\mathbf{p}$. They are differentiably homotopic via the **straight-line homotopy**

$$H(\mathbf{x}, t) = t\, h(\mathbf{x}) + (1-t)\, g(\mathbf{x}),$$

which maps $A \times I$ into A, since $H(\mathbf{x}, t)$ lies on the segment from $h(\mathbf{x}) = \mathbf{x}$ to $g(\mathbf{x}) = \mathbf{p}$. Let $\mathcal{P}$ be as in Theorem 39.2. If f is a 0-form on A, $\mathcal{P}(df) = h^*f - g^*f = f \circ h - f \circ g$; if $df = 0$ then $0 = f(\mathbf{x}) - f(\mathbf{p})$ for all $\mathbf{x}$, so f is constant on A. If ω has order $k > 0$, $d\mathcal{P}\omega + \mathcal{P}d\omega = h^*\omega - g^*\omega$, where $h^*\omega = \omega$ (h is the identity) and $g^*\omega = 0$ (g is constant). If $d\omega = 0$, then $d\mathcal{P}\omega = \omega$, so ω is exact. ■

Theorem 39.4

Let A be a star-convex open set in $\mathbb{R}^n$. Let ω be a closed k-form on A. If $k > 1$, and if η and η_0 are two $k-1$ forms on A with $d\eta = \omega = d\eta_0$, then $\eta = \eta_0 + d\theta$ for some $k-2$ form θ on A. If $k = 1$, and if f and f_0 are two 0-forms on A with $df = \omega = df_0$, then $f = f_0 + c$ for some constant c.

Proof. $d(\eta - \eta_0) = 0$, so $\eta - \eta_0$ is a closed form on A, exact by the Poincaré lemma. Similarly $f - f_0$ is a closed 0-form, hence constant. ■

§40. The deRham Groups of Punctured Euclidean Space

Star-convex open sets are homologically trivial in all dimensions. The simplest set that is not is the punctured euclidean space $\mathbb{R}^n - \mathbf{0}$: earlier we found a closed $(n-1)$-form on $\mathbb{R}^n - \mathbf{0}$ that is not exact. We now give a definitive criterion for when a closed form on $\mathbb{R}^n - \mathbf{0}$ is exact, using vector spaces $H^k(A)$ called the *deRham groups* of A: A is homologically trivial in dimension k iff $H^k(A)$ is trivial.

Definition 40.1 (quotient space)

Let W be a linear subspace of a vector space V. V/W is the set of subsets of V of the form $\mathbf{v} + W = \{\mathbf{v} + \mathbf{w} : \mathbf{w} \in W\}$, called **cosets** of V determined by W. If $\mathbf{v}_1 - \mathbf{v}_2 \in W$ then $\mathbf{v}_1 + W = \mathbf{v}_2 + W$, and if $\mathbf{v}_1 - \mathbf{v}_2 \notin W$ they are disjoint; so V/W is a **partition** of V. With

$$(\mathbf{v}_1 + W) + (\mathbf{v}_2 + W) = (\mathbf{v}_1 + \mathbf{v}_2) + W, \qquad c(\mathbf{v} + W) = (c\mathbf{v}) + W,$$

V/W is a vector space, the **quotient space** of V by W.

The operations are well-defined: if $\mathbf{v}_1 + W = \mathbf{v}_1' + W$ and $\mathbf{v}_2 + W = \mathbf{v}_2' + W$, then $\mathbf{v}_1 - \mathbf{v}_1'$ and $\mathbf{v}_2 - \mathbf{v}_2'$ lie in W, hence so does $(\mathbf{v}_1 + \mathbf{v}_2) - (\mathbf{v}_1' + \mathbf{v}_2')$; similarly for scalar multiplication. If V is finite-dimensional so is V/W; but V/W may be finite-dimensional even when V and W are not.

Definition 40.2 (induced map of quotients)

If $W \subset V$ and $W' \subset V'$ are linear subspaces and $T : V \to V'$ is linear with $T(W) \subset W'$, then $\tilde{T} : V/W \to V'/W'$, $\tilde{T}(\mathbf{v} + W) = T(\mathbf{v}) + W'$, is a well-defined linear transformation, **induced** by T.

Definition 40.3 (deRham group)

Let A be open in $\mathbb{R}^n$. The closed k-forms $\mathcal{C}^k(A)$ and the exact k-forms $\mathcal{E}^k(A)$ are linear subspaces of $\Omega^k(A)$, and $\mathcal{E}^k(A) \subset \mathcal{C}^k(A)$. The **deRham group** of A in dimension k is the quotient vector space

$$H^k(A) = \mathcal{C}^k(A)/\mathcal{E}^k(A).$$

For a closed k-form ω we denote its coset $\omega + \mathcal{E}^k(A)$ by $\{\omega\}$.

$H^k(A)$ is the trivial (zero) vector space iff A is homologically trivial in dimension k. If $g : A \to B$ is a C^∞ map of open sets in $\mathbb{R}^n, \mathbb{R}^m$, then $g^* : \Omega^k(B) \to \Omega^k(A)$ commutes with d, so it carries closed forms to closed forms and exact forms to exact forms, and induces a linear map $g^* : H^k(B) \to H^k(A)$ (we write g^* instead of $\tilde{g}^*$). Two tools from algebraic topology compute these groups: *homotopy equivalence*, and a special case of the *Mayer–Vietoris theorem*.

Theorem 40.1 (Homotopy equivalence theorem)

Let A and B be open sets in $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. Let $g : A \to B$ and $h : B \to A$ be C^∞ maps. If $g \circ h : B \to B$ is differentiably homotopic to the identity map i_B of B, and if $h \circ g : A \to A$ is differentiably homotopic to the identity map i_A of A, then g^* and h^* are linear isomorphisms of the deRham groups.

If $g \circ h = i_B$ and $h \circ g = i_A$, g and h are diffeomorphisms; under the weaker hypotheses of the theorem they are called (differentiable) **homotopy equivalences**.

Proof. If η is a closed k-form on A ($k \geq 0$), Theorem 39.2 implies $(h \circ g)^*\eta - (i_A)^*\eta$ is exact (for $k = 0$ it is zero). So the induced maps satisfy $g^*(h^*(\{\eta\})) = \{\eta\}$: $g^* \circ h^*$ is the identity of $H^k(A)$. Similarly $h^* \circ g^*$ is the identity of $H^k(B)$. The first fact shows g^* maps $H^k(B)$ onto $H^k(A)$ (since $\{\eta\} = g^*(h^*\{\eta\})$); the second shows g^* is one-to-one (if $g^*\{\omega\} = 0$ then $\{\omega\} = h^*(g^*\{\omega\}) = 0$). By symmetry h^* is also a linear isomorphism. ■

Lemma 40.2

Let U and V be open sets in $\mathbb{R}^n$; let $X = U \cup V$; and suppose $A = U \cap V$ is non-empty. Then there exists a C^∞ function $\phi : X \to [0, 1]$ such that ϕ is identically 0 in a neighborhood of $U - A$ and ϕ is identically 1 in a neighborhood of $V - A$.

Proof. Let $\{\phi_i\}$ be a partition of unity on X dominated by $\{U, V\}$, $S_i = \text{Support}\,\phi_i$. Divide the indices into disjoint sets J and K such that $S_i \subset U$ for $i \in J$ and $S_i \subset V$ for $i \in K$ (e.g. $J = \{i : S_i \subset U\}$, K the rest). Let $\phi(\mathbf{x}) = \sum_{i \in K} \phi_i(\mathbf{x})$; by local finiteness ϕ is C^∞ on X. Let $\mathbf{a} \in U - A$. Choose a neighborhood W of $\mathbf{a}$ meeting only finitely many S_i; let D be the union of those among them with $i \in K$. D is closed and does not contain $\mathbf{a}$ (as $S_i \subset V$ for $i \in K$ and $\mathbf{a} \notin V$), so $W - D$ is a neighborhood of $\mathbf{a}$ on which each ϕ_i, $i \in K$, vanishes; thus $\phi = 0$ on $W - D$. Since $1 - \phi(\mathbf{x}) = \sum_{i \in J} \phi_i(\mathbf{x})$, symmetry shows $1 - \phi$ vanishes in a neighborhood of each point of $V - A$. ■

Theorem 40.3 (Mayer–Vietoris—special case)

Let U and V be open sets in $\mathbb{R}^n$ with U and V homologically trivial in all dimensions. Let $X = U \cup V$; suppose $A = U \cap V$ is non-empty. Then $H^0(X)$ is trivial, and for $k \geq 0$, the space $H^{k+1}(X)$ is linearly isomorphic to the space $H^k(A)$.

Proof. For open $B \subset C$ and a k-form η on C, write $\eta|B = j^*\eta$ ($j : B \to C$ the inclusion), the **restriction** of η to B. Since j^* commutes with d, restrictions of closed (exact) forms are closed (exact); and $(\eta|B)|A = \eta|A$ for $A \subset B \subset C$.

Step 1. $H^0(X)$ is trivial. Let f be a closed 0-form on X. $f|U$ and $f|V$ are closed, and U, V are homologically trivial in dimension 0, so $f|U = c_1$, $f|V = c_2$ are constants; as $U \cap V \neq \emptyset$, $c_1 = c_2$ and f is constant.

Step 2. Let $\phi : X \to [0,1]$ be C^∞ with $\phi = 0$ in a neighborhood U' of $U - A$ and $1 - \phi = 0$ in a neighborhood V' of $V - A$ (Lemma 40.2). For $k \geq 0$ define $\delta : \Omega^k(A) \to \Omega^{k+1}(X)$ by

$$\delta(\omega) = \begin{cases} d\phi \wedge \omega & \text{on } A, \\ 0 & \text{on } U' \cup V'. \end{cases}$$

Since $d\phi = 0$ on $U' \cup V'$, $\delta(\omega)$ is well-defined, and since A and $U' \cup V'$ are open with union X, it is C^∞ on X. δ is linear and commutes with d up to sign: $d(\delta(\omega)) = -d\phi \wedge d\omega$ on A and 0 on $U' \cup V'$, i.e. $d\delta(\omega) = -\delta(d\omega)$. So δ carries closed forms to closed forms and exact forms to exact forms ($\delta(d\theta) = -d\delta(\theta)$), and induces a linear map $\tilde{\delta} : H^k(A) \to H^{k+1}(X)$. We show $\tilde{\delta}$ is an isomorphism.

Step 3. $\tilde{\delta}$ is one-to-one: if ω is a closed k-form on A with $\delta(\omega) = d\theta$ for a k-form θ on X, then ω is exact. Define k-forms

$$\omega_1 = \begin{cases} \phi\omega & \text{on } A, \\ 0 & \text{on } U', \end{cases} \qquad \omega_2 = \begin{cases} (1-\phi)\omega & \text{on } A, \\ 0 & \text{on } V', \end{cases}$$

well-defined and C^∞ on U and V. Since $d\omega = 0$, $d\omega_1 = d\phi \wedge \omega = \delta(\omega)$ on A and 0 on U', so $d\omega_1 = \delta(\omega)|U = d\theta|U$; thus $\omega_1 - \theta|U$ is closed on U. Similarly $d\omega_2 = -d\theta|V$ and $\omega_2 + \theta|V$ is closed on V. If $k > 0$, homological triviality gives $(k-1)$-forms η_1, η_2 on U, V with $\omega_1 - \theta|U = d\eta_1$, $\omega_2 + \theta|V = d\eta_2$. Restricting to A and adding,

$$\phi\omega + (1-\phi)\omega = \omega_1|A + \omega_2|A = d(\eta_1|A + \eta_2|A),$$

so ω is exact on A. If $k = 0$, there are constants c_1, c_2 with $\omega_1 - \theta|U = c_1$, $\omega_2 + \theta|V = c_2$, and $\phi\omega + (1-\phi)\omega = c_1 + c_2$ on A; so ω is constant, i.e. exact.

Step 4. $\tilde{\delta}$ is onto: given a closed $(k+1)$-form η on X, we find a closed k-form ω on A with $\eta - \delta(\omega)$ exact. $\eta|U$ and $\eta|V$ are closed, so there are k-forms θ_1, θ_2 on U, V with $d\theta_1 = \eta|U$, $d\theta_2 = \eta|V$. Let $\omega = \theta_1|A - \theta_2|A$; it is closed since $d\omega = \eta|A - \eta|A = 0$. Define a k-form θ on X by

$$\theta = \begin{cases} (1-\phi)\theta_1 + \phi\theta_2 & \text{on } A, \\ \theta_1 & \text{on } U', \\ \theta_2 & \text{on } V', \end{cases}$$

well-defined and C^∞. We show $d\theta = \eta - \delta(\omega)$. On A:

$$d\theta|A = [-d\phi \wedge (\theta_1|A) + (1-\phi)(d\theta_1|A)] + [d\phi \wedge (\theta_2|A) + \phi(d\theta_2|A)] = \eta|A + d\phi \wedge (-\omega) = \eta|A - \delta(\omega)|A.$$

On U' and V': $d\theta|U' = d\theta_1|U' = \eta|U' = \eta|U' - \delta(\omega)|U'$ and similarly on V', since $\delta(\omega)$ vanishes on $U' \cup V'$. Hence $d\theta = \eta - \delta(\omega)$. ∎

Theorem 40.4

Let $n \geq 1$. Then

$$\dim H^k(\mathbb{R}^n - \mathbf{0}) = \begin{cases} 0 & \text{for } k \neq n-1, \\ 1 & \text{for } k = n-1. \end{cases}$$

Proof.

Step 1. $n = 1$. Let $A = \mathbb{R}^1 - 0 = A_0 \cup A_1$, with A_0 the negative and A_1 the positive reals. If ω is a closed k-form on A with $k > 0$, $\omega|A_0$ and $\omega|A_1$ are closed; A_0, A_1 are star-convex, so there are

$(k-1)$-forms η_i on A_i with $d\eta_i = \omega|A_i$. Let $\eta = \eta_0$ on A_0 and η_1 on A_1; then η is C^∞ on A and $d\eta = \omega$. For $k = 0$: let $f_0 = 0$ on A_0 and $f_0 = 1$ on A_1. f_0 is a closed 0-form that is not constant, hence not exact. $\{f_0\}$ is a basis of $H^0(A)$: if f is a closed 0-form on A, $f|A_0 = c_0$ and $f|A_1 = c_1$ are constants, so $f = c_1 f_0 + c_0(1 - f_0)$, i.e. $f = (c_1 - c_0)f_0 + c_0$, and $\{f\} = (c_1 - c_0)\{f_0\}$.

Step 2. If B is open in $\mathbb{R}^n$, then $B \times \mathbb{R}$ is open in $\mathbb{R}^{n+1}$ and $\dim H^k(B) = \dim H^k(B \times \mathbb{R})$ for all k. Let $g : B \to B \times \mathbb{R}$, $g(\mathbf{x}) = (\mathbf{x}, 0)$, and $h : B \times \mathbb{R} \to B$, $h(\mathbf{x}, s) = \mathbf{x}$. Then $h \circ g$ is the identity of B, and $g \circ h$ is differentiably homotopic to the identity of $B \times \mathbb{R}$ via the straight-line homotopy $H((\mathbf{x}, s), t) = t(\mathbf{x}, s) + (1 - t)(\mathbf{x}, 0) = (\mathbf{x}, st)$. Apply Theorem 40.1.

Step 3. Assume the theorem for n $(n \geq 1)$ and prove it for $n + 1$. Let

$$U = \mathbb{R}^{n+1} - \{(0, \ldots, 0, t) : t \geq 0\}, \qquad V = \mathbb{R}^{n+1} - \{(0, \ldots, 0, t) : t \leq 0\},$$

so U is $\mathbb{R}^{n+1}$ minus the half-line $\mathbf{0} \times \mathbb{H}^1$ and V is $\mathbb{R}^{n+1}$ minus the half-line $\mathbf{0} \times \mathbb{L}^1$. Then $A = U \cap V = \mathbb{R}^n \times \mathbb{R}$ minus the line $\mathbf{0} \times \mathbb{R}$, i.e. $A = (\mathbb{R}^n - \mathbf{0}) \times \mathbb{R}$, and $X = U \cup V = \mathbb{R}^{n+1} - \mathbf{0}$. U is star-convex with respect to $\mathbf{p} = (0, \ldots, 0, -1)$ and V with respect to $\mathbf{q} = (0, \ldots, 0, 1)$. By Theorem 40.3, $H^0(X)$ is trivial and $\dim H^{k+1}(X) = \dim H^k(A)$ for $k \geq 0$. By Step 2, $\dim H^k(A) = \dim H^k(\mathbb{R}^n - \mathbf{0})$, which by the induction hypothesis is 0 if $k \neq n - 1$ and 1 if $k = n - 1$. So $\dim H^{k+1}(\mathbb{R}^{n+1} - \mathbf{0})$ is 0 if $k + 1 \neq n$ and 1 if $k + 1 = n$. ■

Restated in terms of forms:

Theorem 40.5

Let $A = \mathbb{R}^n - \mathbf{0}$, with $n \geq 1$.
(a) If $k \neq n - 1$, then every closed k-form on A is exact on A.
(b) There is a closed $n - 1$ form η_0 on A that is not exact. If η is any closed $n - 1$ form on A, then there is a unique scalar c such that $\eta - c\eta_0$ is exact.

Proof [Supplement]. (a) $H^k(A) = 0$.

(b) $H^{n-1}(A)$ has dimension 1; take η_0 with $\{\eta_0\}$ a basis. Then $\{\eta\} = c\{\eta_0\}$ for a unique c. ■

This guarantees a closed non-exact $(n-1)$-form on $\mathbb{R}^n - \mathbf{0}$ without giving a formula. An explicit one is

$$\eta_0 = \sum_{i=1}^{n} (-1)^{i-1} f_i \, dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n, \qquad f_i(\mathbf{x}) = x_i / \|\mathbf{x}\|^n.$$

Why η_0 is closed but not exact [Supplement]

$\eta_0 = \beta_{n-1} G$ for the vector field $G(\mathbf{x}) = (\mathbf{x}; \mathbf{x}/\|\mathbf{x}\|^n)$, so $d\eta_0 = (\operatorname{div} G)\, dx_1 \wedge \cdots \wedge dx_n$ (Theorem 31.1). With $r = \|\mathbf{x}\|$, $D_i(x_i/r^n) = 1/r^n - n x_i^2/r^{n+2}$, so $\operatorname{div} G = n/r^n - n r^2/r^{n+2} = 0$. On $S^{n-1} = \partial B^n$ with the induced orientation, the corresponding normal is the outward one $N(\mathbf{x}) = (\mathbf{x}; \mathbf{x})$ (Lemma 38.7), and $\langle G, N \rangle = \|\mathbf{x}\|^2/\|\mathbf{x}\|^n = 1$ there; by Lemma 38.5, $\int_{S^{n-1}} \eta_0 = \mathrm{v}(S^{n-1}) \neq 0$. If $\eta_0 = d\theta$, Stokes' theorem would give $\int_{S^{n-1}} \eta_0 = \int_{\partial S^{n-1}} \theta = 0$ (S^{n-1} has empty boundary).

Theorem 40.6

Let $A = \mathbb{R}^n - \mathbf{0}$, with $n > 1$. If η is a closed $n - 1$ form in A, then η is exact in A if and only if

$$\int_{S^{n-1}} \eta = 0.$$

Proof. If η is exact, its integral over S^{n-1} is 0 by Stokes' theorem. Conversely suppose the integral is zero. By Theorem 40.5 there is a unique c with $\eta - c\eta_0$ exact. Then, by Stokes' theorem, $\int_{S^{n-1}} \eta = c \int_{S^{n-1}} \eta_0$. Since $\int_{S^{n-1}} \eta_0 \neq 0$, $c = 0$, and η is exact. ■

Selected exercises

Motivation. These exercises from Munkres' §39–§40 show how the two computing tools of the chapter, invariance under diffeomorphisms and homotopies and the Mayer–Vietoris argument, are used beyond the single example $\mathbb{R}^n - \mathbf{0}$.

Exercise 39.2 (diffeomorphism invariance; a non-star-convex trivial set)

(a) Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^n$, of class C^∞. Show that if A is homologically trivial in dimension k, so is B.
(b) Find an open set in $\mathbb{R}^2$ that is not star-convex but is homologically trivial in every dimension.

Solution.

(a) Let $h = g^{-1} : B \to A$; it is of class C^∞, and $g \circ h = i_B$.

Step 1. ($k = 0$.) Let f be a closed 0-form on B. By Theorem 32.3, $d(f \circ g) = d(g^* f) = g^*(df) = 0$, so $f \circ g$ is a closed 0-form on A, hence equal to a constant c. Then $f = (f \circ g) \circ h = c$ on B.

Step 2. ($k > 0$.) Let ω be a closed k-form on B. Then $d(g^*\omega) = g^*(d\omega) = 0$, so $g^*\omega = d\theta$ for some $(k-1)$-form θ on A. By Theorem 32.1(3) and Theorem 32.3,

$$\omega = (g \circ h)^*\omega = h^*(g^*\omega) = h^*(d\theta) = d(h^*\theta),$$

so ω is exact on B.

(b) Let $R_1 = (0,3) \times (0,1)$, $R_2 = (0,1) \times (0,3)$, $R_3 = (2,3) \times (0,3)$ and let $X = R_1 \cup R_2 \cup R_3$ (a "U"-shaped region).

Step 1. (X is homologically trivial.) Put $U = R_1 \cup R_2$ and $V = R_3$. The set U is star-convex with respect to $\mathbf{p} = (\frac{1}{2}, \frac{1}{2})$: each R_i is convex and contains $\mathbf{p}$, so the segment from $\mathbf{p}$ to a point of R_i lies in $R_i \subset U$. V is convex, and $U \cap V = (2,3) \times (0,1)$ is convex and non-empty. By the Poincaré lemma (Theorem 39.3, which includes $k = 0$ since exact 0-forms are the constants), U, V and $U \cap V$ are homologically trivial in all dimensions. Theorem 40.3 gives $H^0(X) = 0$ and $H^{k+1}(X) \cong H^k(U \cap V) = 0$ for $k \geq 0$.

Step 2. (X is not star-convex.) The reflection $\sigma(x, y) = (3 - x, y)$ is affine, maps segments to segments, and satisfies $\sigma(R_1) = R_1$, $\sigma(R_2) = R_3$, so $\sigma(X) = X$. Hence it suffices to show that X is not star-convex with respect to any $\mathbf{p} = (p_1, p_2) \in X$ with $p_1 \leq \frac{3}{2}$. Let $\mathbf{b} = (\frac{5}{2}, \frac{5}{2}) \in R_3$ and $\mathbf{c}(s) = \mathbf{p} + s(\mathbf{b} - \mathbf{p})$. The first coordinate of $\mathbf{c}(s)$ equals 2 at

$$s_0 = \frac{2 - p_1}{5/2 - p_1} \in \left[\tfrac{1}{2}, \tfrac{4}{5}\right) \qquad (0 < p_1 \leq \tfrac{3}{2}).$$

Its second coordinate there is $y_0 = (1 - s_0)p_2 + \frac{5}{2}s_0 \geq \frac{5}{2}s_0 \geq \frac{5}{4}$, since $p_2 > 0$. The point $(2, y_0)$ is not in R_1 (as $y_0 \geq 1$), not in R_2 (as $2 \geq 1$) and not in R_3 (as $2 \not> 2$). So the segment from $\mathbf{p}$ to $\mathbf{b}$ leaves X. ■

Exercise 39.3 (dimension 0 and connectedness)

Let A be a non-empty open set in $\mathbb{R}^n$. Show that A is homologically trivial in dimension 0 if and only if A is connected. (Connectedness is Definition 4.4.)

Solution.

Step 1. (Not connected ⇒ not trivial.) Suppose $A = A_0 \cup A_1$ with A_0, A_1 disjoint, non-empty and open. Let $f = 0$ on A_0 and $f = 1$ on A_1. Every point of A has a neighborhood on which f is

constant, so f is C^∞ and $df = 0$ by Theorem 30.3. But f is not constant, so it is a closed 0-form that is not exact.

Step 2. (Closed 0-forms are constant on segments.) Let $df = 0$ on A; by Theorem 30.3, $D_j f = 0$ for all j. If the segment from $\mathbf{a}$ to $\mathbf{b}$ lies in A, then $\phi(t) = f(\mathbf{a} + t(\mathbf{b} - \mathbf{a}))$ satisfies $\phi'(t) = Df(\mathbf{a} + t(\mathbf{b} - \mathbf{a})) \cdot (\mathbf{b} - \mathbf{a}) = 0$ on $[0, 1]$, so $f(\mathbf{b}) = \phi(1) = \phi(0) = f(\mathbf{a})$ by the mean value theorem. Hence $f(\mathbf{x}) = f(\mathbf{p})$ whenever $\mathbf{x}$ can be joined to $\mathbf{p}$ by a broken-line path in A.

Step 3. (Connected $\Rightarrow$ trivial.) Fix $\mathbf{p} \in A$ and let S be the set of points of A that can be joined to $\mathbf{p}$ by a broken-line path in A. If $\mathbf{x} \in A$, choose an open ball $B \subset A$ centered at $\mathbf{x}$; every $\mathbf{y} \in B$ is joined to $\mathbf{x}$ by a segment in B. If $\mathbf{x} \in S$, appending this segment shows $B \subset S$; if $\mathbf{x} \notin S$, then no $\mathbf{y} \in B$ lies in S (otherwise $\mathbf{x}$ would be joined to $\mathbf{p}$ through $\mathbf{y}$), so $B \subset A - S$. Thus S and $A - S$ are open, disjoint, with union A, and $\mathbf{p} \in S$. Since A is connected, $S = A$, and by Step 2 every closed 0-form on A is constant. ■

Exercise 40.4 (what the Mayer–Vietoris argument really needs)

Let U and V be open sets in $\mathbb{R}^n$; let $X = U \cup V$ and assume $A = U \cap V$ is non-empty. Let $\tilde{\delta} : H^k(A) \to H^{k+1}(X)$ be the map constructed in the proof of Theorem 40.3. What hypotheses on the deRham groups of U and V ensure that
(a) $\tilde{\delta}$ is one-to-one?
(b) the image of $\tilde{\delta}$ is all of $H^{k+1}(X)$?
(c) $H^0(X)$ is trivial?

Solution. The construction of ϕ, δ and $\tilde{\delta}$ (Lemma 40.2 and Step 2 of the proof of Theorem 40.3) uses no hypothesis on U and V, so $\tilde{\delta}$ is always a well-defined linear map. We check which facts each of Steps 1, 3 and 4 uses.

(a) It suffices that $H^k(U) = 0$ and $H^k(V) = 0$. Let ω be a closed k-form on A with $\tilde{\delta}\{\omega\} = 0$, i.e. $\delta(\omega) = d\theta$ for a k-form θ on X. With ω_1, ω_2 as in Step 3, the computation there gives $d(\omega_1 - \theta|U) = 0$ and $d(\omega_2 + \theta|V) = 0$, without any hypothesis. If $H^k(U) = H^k(V) = 0$, these closed k-forms are exact: for $k > 0$, $\omega_1 - \theta|U = d\eta_1$ and $\omega_2 + \theta|V = d\eta_2$, and adding the restrictions to A gives

$$\omega = \phi\omega + (1 - \phi)\omega = d(\eta_1|A + \eta_2|A);$$

for $k = 0$ the two forms are constants c_1, c_2 and $\omega = c_1 + c_2$. In both cases $\{\omega\} = 0$.

(b) It suffices that $H^{k+1}(U) = 0$ and $H^{k+1}(V) = 0$. Let η be a closed $(k+1)$-form on X. The only place where Step 4 uses a hypothesis is the existence of k-forms θ_1, θ_2 on U, V with $d\theta_1 = \eta|U$, $d\theta_2 = \eta|V$, which is exactly the exactness of the closed forms $\eta|U$, $\eta|V$. Given these, $\omega = \theta_1|A - \theta_2|A$ is closed and the form θ of Step 4 satisfies $d\theta = \eta - \delta(\omega)$, so $\{\eta\} = \tilde{\delta}\{\omega\}$.

(c) It suffices that $H^0(U) = 0$ and $H^0(V) = 0$, i.e. (Exercise 39.3) that U and V are connected. Indeed, a closed 0-form f on X then has constant restrictions $f|U = c_1$, $f|V = c_2$, and evaluating at a point of $A \neq \emptyset$ gives $c_1 = c_2$, so f is constant.

Assuming (a) and (b) for every $k \geq 0$ together with (c) is exactly the hypothesis "U and V homologically trivial in all dimensions" of Theorem 40.3. ■

Exercise 40.5 (two punctures)

Let $\mathbf{p}$ and $\mathbf{q}$ be two distinct points of $\mathbb{R}^n$; let $n \geq 1$. Show that

$$\dim H^k(\mathbb{R}^n - \mathbf{p} - \mathbf{q}) = \begin{cases} 0 & \text{if } k \neq n-1, \\ 2 & \text{if } k = n-1. \end{cases}$$

Solution.

Step 1. (Invariance.) If $T:\mathbb{R}^n\to\mathbb{R}^n$ is a diffeomorphism and B is open, then $T|B: B\to T(B)$ and $T^{-1}|T(B)$ are C^∞ maps whose composites are identity maps; these are differentiably homotopic to themselves (take $H(\mathbf{x},t)=\mathbf{x}$), so Theorem 40.1 gives $H^k(B)\cong H^k(T(B))$. Take $T(\mathbf{x})=Q(\mathbf{x}-\mathbf{p})$ with Q orthogonal and $Q(\mathbf{q}-\mathbf{p})=d\,\mathbf{e}_1$, $d=\|\mathbf{q}-\mathbf{p}\|$ (complete $(\mathbf{q}-\mathbf{p})/d$ to an orthonormal basis). Then $T(\mathbb{R}^n-\mathbf{p}-\mathbf{q})=\mathbb{R}^n-\mathbf{0}-d\mathbf{e}_1$. So the dimensions depend only on n, and in $\mathbb{R}^{n+1}$ we may place the two points in $\mathbb{R}^n\times 0$.

Step 2. ($n=1$.) Let $p<q$ and $A=\mathbb{R}-p-q=A_1\cup A_2\cup A_3$ with $A_1=(-\infty,p)$, $A_2=(p,q)$, $A_3=(q,\infty)$. If $k\geq 1$ and ω is a closed k-form on A, the Poincaré lemma gives $(k-1)$-forms η_i on the intervals A_i with $d\eta_i=\omega|A_i$; the form η equal to η_i on A_i is C^∞ (the A_i are open and disjoint) and $d\eta=\omega$. So $H^k(A)=0$ for $k\geq 1$. For $k=0$, let χ_i be the function equal to 1 on A_i and 0 on the other two intervals; it is a closed 0-form. A closed 0-form f is constant, say c_i, on each A_i (Theorem 39.3), so

$$f=(c_1-c_3)\chi_1+(c_2-c_3)\chi_2+c_3 \qquad \text{and} \qquad \{f\}=(c_1-c_3)\{\chi_1\}+(c_2-c_3)\{\chi_2\}.$$

If $a\chi_1+b\chi_2$ is constant, its values a, b, 0 on A_1, A_2, A_3 coincide, so $a=b=0$. Hence $\{\chi_1\},\{\chi_2\}$ is a basis and $\dim H^0(A)=2$, as claimed for $n=1$.

Step 3. (Two slit spaces.) Write points of $\mathbb{R}^{n+1}$ as $(\mathbf{x},t)$. For distinct $\mathbf{p},\mathbf{q}\in\mathbb{R}^n$ let

$$U=\mathbb{R}^{n+1}-\{\mathbf{p},\mathbf{q}\}\times[0,\infty), \qquad V=\mathbb{R}^{n+1}-\{\mathbf{p},\mathbf{q}\}\times(-\infty,0].$$

We show U is homologically trivial in all dimensions. Let $\ell(\mathbf{x})=\langle\mathbf{q}-\mathbf{p},\mathbf{x}\rangle$; then $\ell(\mathbf{q})-\ell(\mathbf{p})=\|\mathbf{q}-\mathbf{p}\|^2>0$, so we can choose $c_1<c_2$ with $\ell(\mathbf{p})<c_1<c_2<\ell(\mathbf{q})$. Let

$$W_1=\{(\mathbf{x},t):\ell(\mathbf{x})<c_2\}-\mathbf{p}\times[0,\infty), \qquad W_2=\{(\mathbf{x},t):\ell(\mathbf{x})>c_1\}-\mathbf{q}\times[0,\infty).$$

They are open. The ray $\mathbf{q}\times[0,\infty)$ does not meet the half-space $\ell<c_2$, so $W_1\subset U$; likewise $W_2\subset U$. A point of U with $\ell(\mathbf{x})<c_2$ lies in W_1, and one with $\ell(\mathbf{x})\geq c_2>c_1$ lies in W_2; so $U=W_1\cup W_2$. Also $W_1\cap W_2=\{(\mathbf{x},t):c_1<\ell(\mathbf{x})<c_2\}$ is convex and non-empty.

W_1 is star-convex with respect to $(\mathbf{p},-1)$. Let $(\mathbf{x},t)\in W_1$ and consider $\mathbf{c}(s)=((1-s)\mathbf{p}+s\mathbf{x},\ -(1-s)+st)$, $s\in[0,1]$. Its ℓ-value $(1-s)\ell(\mathbf{p})+s\ell(\mathbf{x})$ is $<c_2$. If $\mathbf{c}(s)$ were on the ray $\mathbf{p}\times[0,\infty)$, then $s(\mathbf{x}-\mathbf{p})=\mathbf{0}$; $s=0$ is impossible (last coordinate -1), and if $\mathbf{x}=\mathbf{p}$ then $t<0$ because $(\mathbf{x},t)\notin\mathbf{p}\times[0,\infty)$, so the last coordinate $-(1-s)+st$ is negative. Hence the segment lies in W_1. Similarly W_2 is star-convex with respect to $(\mathbf{q},-1)$.

By the Poincaré lemma W_1, W_2, $W_1\cap W_2$ are homologically trivial in all dimensions, so Theorem 40.3 gives $H^0(U)=0$ and $H^{k+1}(U)\cong H^k(W_1\cap W_2)=0$. The map $(\mathbf{x},t)\mapsto(\mathbf{x},-t)$ is a diffeomorphism of $\mathbb{R}^{n+1}$ carrying U onto V, so V is also homologically trivial in all dimensions by Exercise 39.2(a).

Step 4. (Induction.) Assume the result for $n\geq 1$ and all pairs of points. By Step 1 we may take the two points of $\mathbb{R}^{n+1}$ to be $(\mathbf{p},0)$ and $(\mathbf{q},0)$ with $\mathbf{p}\neq\mathbf{q}$ in $\mathbb{R}^n$. With U, V as in Step 3, a point $(\mathbf{x},t)$ lies outside $U\cup V$ iff $\mathbf{x}\in\{\mathbf{p},\mathbf{q}\}$ and $t\geq 0$ and $t\leq 0$; hence

$$X=U\cup V=\mathbb{R}^{n+1}-(\mathbf{p},0)-(\mathbf{q},0), \qquad A=U\cap V=(\mathbb{R}^n-\mathbf{p}-\mathbf{q})\times\mathbb{R}\neq\emptyset.$$

By Step 3 and Theorem 40.3, $H^0(X)=0$ and $\dim H^{k+1}(X)=\dim H^k(A)$ for $k\geq 0$. By Step 2 of the proof of Theorem 40.4, $\dim H^k(A)=\dim H^k(\mathbb{R}^n-\mathbf{p}-\mathbf{q})$, which is 2 for $k=n-1$ and 0 otherwise. Hence $\dim H^j(X)$ is 2 for $j=n=(n+1)-1$ and 0 for all other $j\geq 0$ (for $j=0$ note $0\neq n$). This is the claim for $n+1$. ∎